

Introduction to Oral & maxillofacial surgery

Oral & maxillofacial surgery

- ▶ Specialty of dentistry which includes the diagnosis, surgical treatment of diseases, injuries and defects involving the hard and soft tissues of the oral and maxillofacial region.



Scope of OMFS

1. Dentoalveolar surgery.
2. Surgery for cranio-maxillofacial trauma, abnormalities and oncology.
3. Diagnosis and treatment of congenital maxillofacial abnormalities including cleft lip & palate .
4. Management of oral and maxillofacial trauma.
5. Diagnosis and treatment of chronic facial pain disorders •

Scope of OMFS (cont.)

6. Surgical and non-surgical managements of diseases and disorders of temporomandibular joint (TMJ).

7. Cosmetic surgery limited to the head and neck.

8. Preprosthetic surgery.

9. Orthognathic surgery.

10. Implantology.

11. Surgical and non-surgical managements of salivary gland diseases.

Anesthesia

- ▶ **Local anesthesia** : loss of sensation without loss of consciousness.
- ▶ **General anesthesia** : loss of consciousness and loss of sensation.
- ▶ Reusable syringe for injection



Informed consent

- ▶ The process by which a patient learns about and understands :
 - ▶ Risks.
 - ▶ Benefits.
 - ▶ Complications.
 - ▶ alternatives, including no treatment
- ▶ Informed consent generally requires the patient or responsible party to sign a statement confirming that they understand the risks and benefits of the procedure or treatment.

Types of surgery

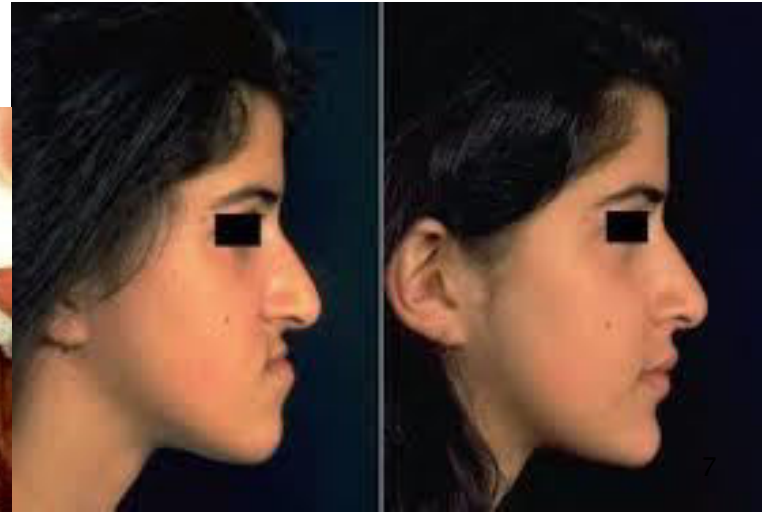
- ▶ **Minor surgery :**

Are usually performed under local anesthesia

- ▶ **Major surgery :**

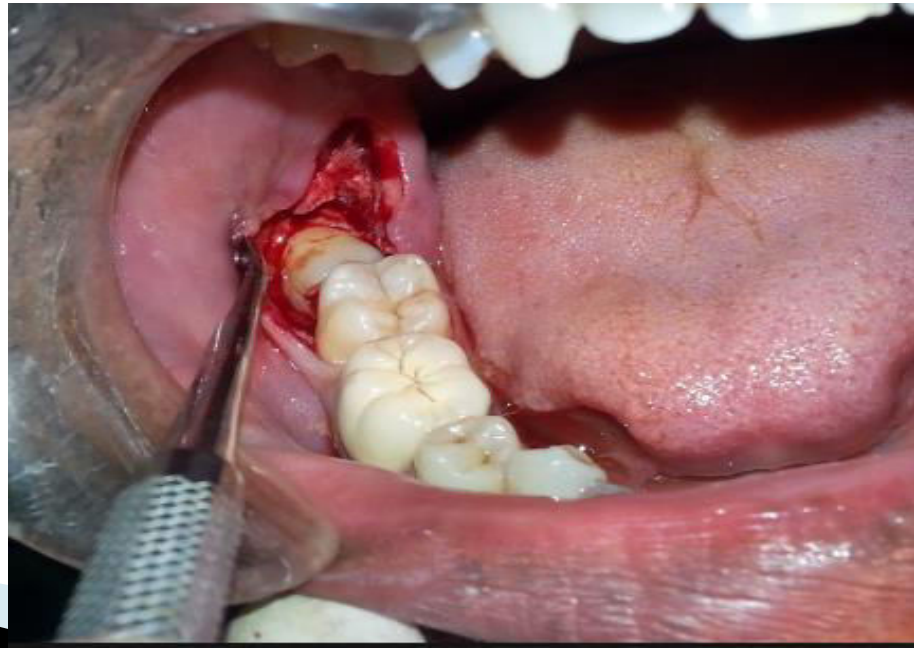
Are usually performed under general anesthesia such as :

- Surgical oncology(removal of oral cancer)
- Maxillofacial trauma.
- Orthognathic surgery.



Minor surgery

- ▶ **A-Surgical extraction** : involve the removal of teeth that cannot be easily accessed, either because they have broken under the gum line or because they have not fully erupted.
- ▶ Are usually performed under local anesthesia
- ▶ Such as impacted wisdom or canine
- ▶ Third molars are the most frequently impacted teeth.



B–Surgical exposure of impacted teeth

- ▶ Performed under local anesthesia.
- ▶ The un-erupted canine will then be exposed by lifting the gum, and guided into place using a special bracket.
- ▶ Permanent maxillary canines are the second most frequently impacted teeth.



The order of frequency of impacted teeth is as follow:

- ▶ 1. mandibular 3rd molar
- ▶ 2. maxillary 3rd molar
- ▶ 3. maxillary cuspid
- ▶ 4. mandibular cuspid
- ▶ 5. Mandibular premolar
- ▶ 6. maxillary premolar
- ▶ 7. maxillary central and lateral incisor

C– Surgical Endodontics

Apicoectomy : surgical procedure to remove root tip (apical portion) Together with the pathological periapical Tissues.

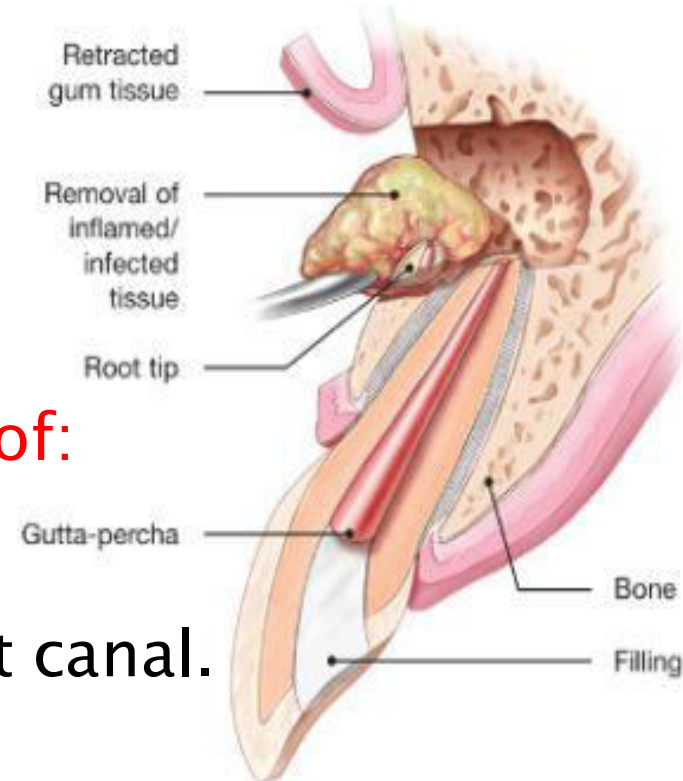
▶ Usually performed after failure of endodontic therapy.

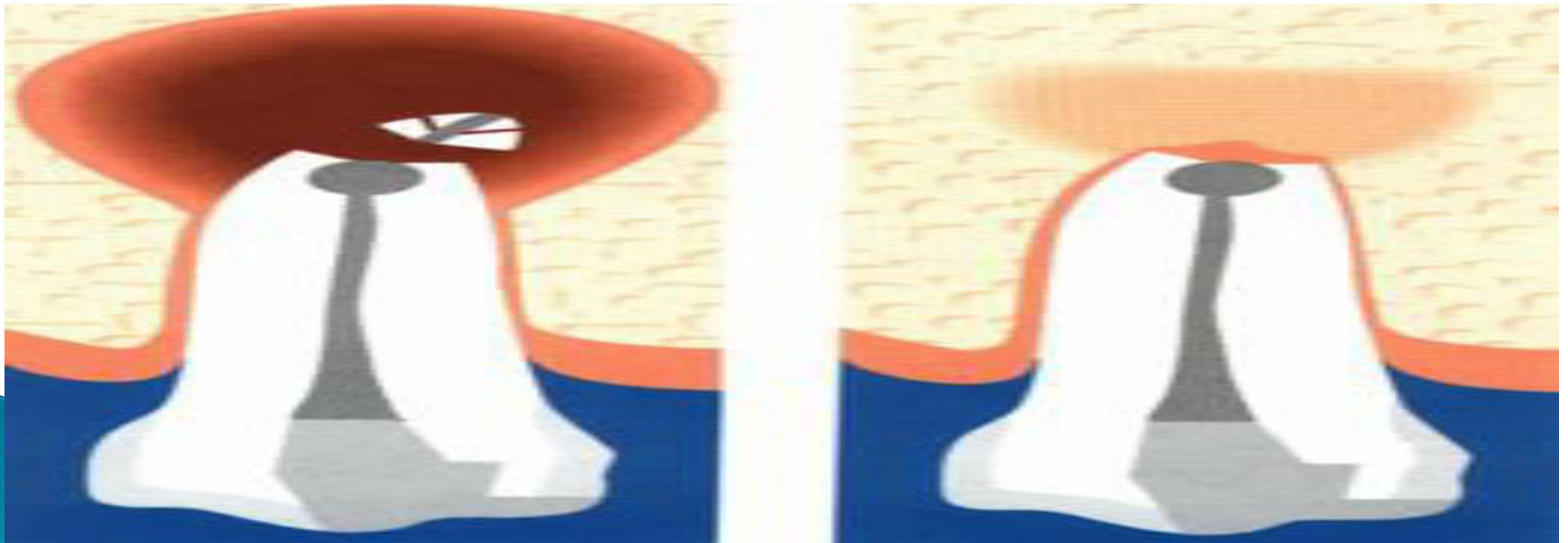
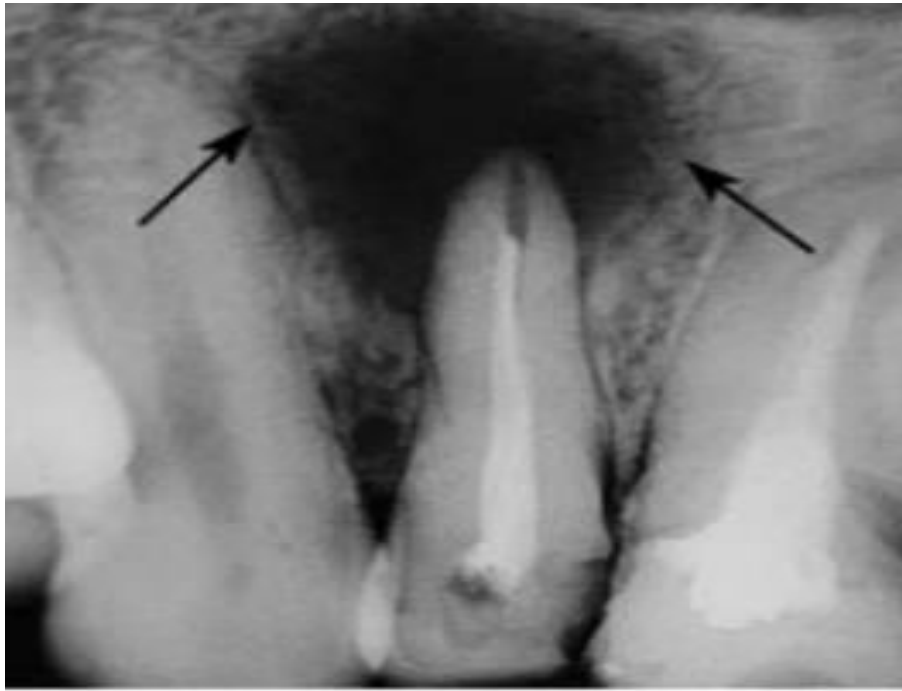
▶ This is usually necessitated when:

1 –A conventional satisfactory root canal therapy had failed.

2 –Unsatisfactory endodontic therapy, which cannot be repeated because of:

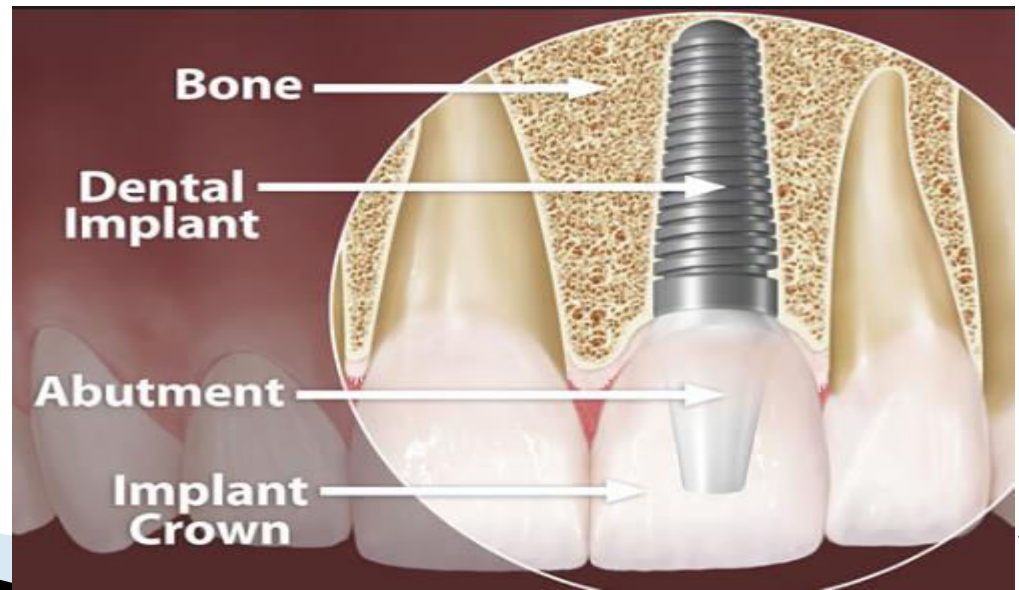
- Severely curved root canals.
- Presence of posts in root canal.
- Breakage of small instrument in root canal.





D. Implantology & Preprosthetic surgery

- ▶ **The implant** : is an artificial dental root , usually made of titanium that is surgically inserted into the jaw bone & that can be used to support a prosthesis.
- ▶ **Dental implant** is often the treatment of choice and is also one of the most conservative treatments for tooth replacement.



Bone graft

- ▶ **Bone graft:**

A material from patient's own body, an artificial, synthetic, or natural substitute used in a surgical procedure that replaces missing bone.

- ▶ As the teeth are lost continuous bone resorption will follow
- ▶ This will interfere with the placement of dental implant, so a bone graft may be indicated to increase the height and/or width of the alveolar bone.

Types of bone graft materials

- ▶ **Autogenous graft (autograft) :** (from the same individual)
- ▶ **Allograft :** (from the same species – cadaver)
- ▶ **Xenograft :** (from different species–animals)
- ▶ **Alloplast :** (artificial bone)

Preprosthetic surgery

- ▶ Surgical procedure that can modify the oral anatomy(soft or hard tissues) to facilitate the fabrication of prosthesis.

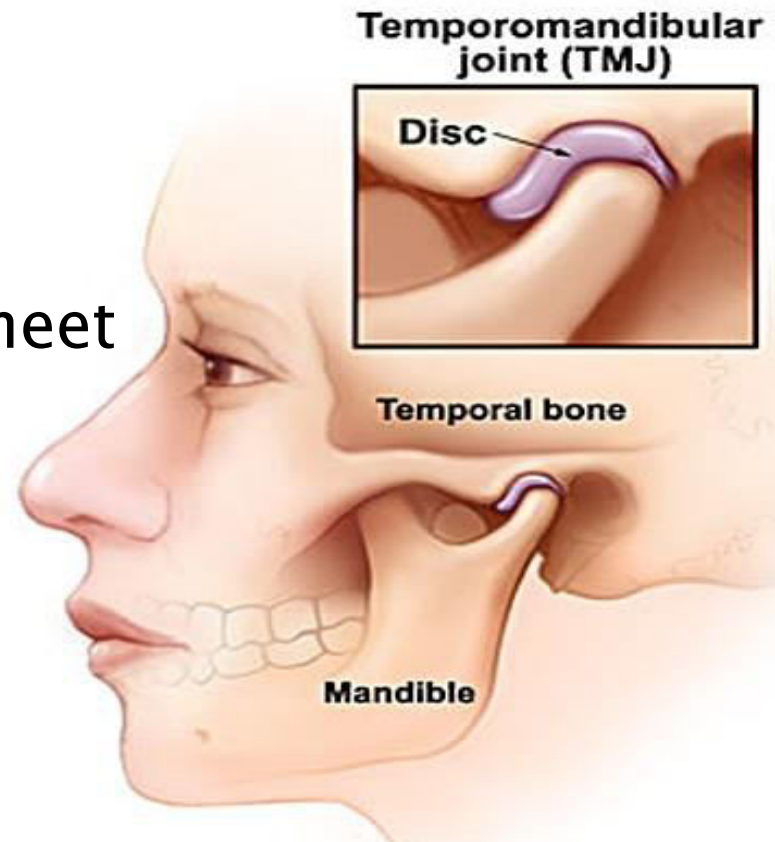


Major surgery

- ▶ Usually performed under general anesthesia in operation room such as:

1 –TMJ (Temporomandibular joint) tumors and disorders:

- ▶ **TMJ:** A small joint located in the front of the ear where the skull and the mandible meet



2- Maxillofacial trauma



3. Surgical oncology

- ▶ Benign or malignant tumors



Fig. 1. Typical presentation of oral malignant melanoma. Note the surrounding areas of pigmentation.

4. Orthognathic surgery

- ▶ For Function & Aesthetic



THANK YOU